

Hein Htut Zaw

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EDUCATION

The State University of New York, Korea - Stony Brook NY

BSc in Computer Science (Specialization: Data Science & Artificial Intelligence)

Graduation: August 2025

GPA: 3.86/4.00, Summa Cum Laude

WORK EXPERIENCE

Machine Learning Engineer - SPOit

February 2026 - Present

- Built an end-to-end football analytics pipeline for automated highlight extraction using DeepEIoU, XFeat, LightGlue, and 2D/3D CNNs and Vision Transformers (ViTs) for player tracking, camera registration, and event classification.
- Implemented multi-camera stitching and homography-based field mapping, enabling unified tactical analysis across 4 camera feeds and improved tracking data accuracy by 20%
- Optimized batch processing and multithreaded inference pipeline, using OpenCV, PyTorch, GPU acceleration, and FP16/TensorRT, reducing processing latency by 40% and increasing throughput to 150fps
- Architected and executed the full ML lifecycle end-to-end, including data collection, labeling, preprocessing pipelines, model fine-tuning, validation, and evaluation tooling, improving model reliability and iteration speed
- Led a cross-functional engineering team of 3 software engineers, defining technical roadmaps, assigning tasks, conducting code reviews, and coordinating delivery of production ML systems.

Software Engineer Intern - SPOit

August 2025 - January 2026

- Published production mobile application supporting video capture and upload workflows.
- Built full-stack architecture using Flask (backend), Expo React Native (frontend), Supabase (DB), implementing staging and production environments with CI/CD pipelines, reducing development iteration time by 40%
- Integrated Docker-based deployments on Google Cloud Platform, enabling transcoding, video transcoding, storage, and backend hosting, reducing deployment time from 1 hour to 15 minutes
- Architected a scalable event-driven media processing pipeline on Google Cloud Platform using Dockerized microservices, cloud storage, and asynchronous job processing for video ingestion and transcoding.

Undergraduate Research Assistant - SUNY Korea

February 2025 - August 2025

- Investigated monocular camera calibration using single-image depth estimation and KITTI datasets.
- Evaluated ML-Depth Pro and Depth Anything against LiDAR ground truth and ICP-based alignment methods.

SKILLS AND CERTIFICATIONS

Programming Languages: Python, C, C++, Java, JavaScript, SQL

Frameworks/Tools: ROS2, Pytorch, Scikit-learn, TensorFlow, Pandas, Numpy, Git, GCP, Docker, DataBricks

Languages: English (IELTS Academic 8), Korean (TOPIK 4)

PROJECTS

[Semantic Book Recommender](#) | ChromaDB, Hugging Face, LangChain, Gradio

June 2026

- Built a semantic book recommendation system using vector embeddings and similarity search over **5,200+ titles**
- Implemented embedding generation and retrieval pipelines with **OpenAI embeddings** and **ChromaDB**
- Applied **BART-MNLI** zero-shot classification to infer missing genres (**77.8% accuracy**) and transformer-based emotion scoring for personalized ranking
- Developed a two-stage retrieval pipeline and deployed an interactive **Gradio** application for real-time recommendations

References available upon request